D 31526 **(Pages : 4)** **Name.....**

Reg. No.....

THIRD SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2022

B.Com./B.B.A.

BCM 3A 11/BBA 3A 11—BASIC NUMERICAL SKILLS

(2017—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A

Answer **all** *questions.*
Each question carries 1 *mark.*

I. Choose the correct answer :

1 Tabulation is the presentation of data in :

a) Groups. b) Rows.

c) Columns. d) Rows and columns.

2 Statistics deals with :

a) Qualitative information. b) Quantitative information.

c) Both. d) None.

3 Skewness refers to :

a) Symmetry. b) Asymmetry.

c) Flatness. d) Peakedness.

4 ———————— Index number is called ideal Index number.

a) Kelley's. b) Paasche's.

c) Laspeyer's d) Fisher's.

5 Mean of 3 items is 25, two of them are 25 and 35. What is the other ?

a) 40. b) 30.

c) 15. d) 60.

Turn over

II. Fill in the blanks :

6 If $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, 3A is ————————.

7 If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, $A \cap B$ = ————————.

8 Simple interest for Rs. 10,000 at the rate of 10 % per annum for 2 years is ————————.

9 If $2x + 3 = 7$, value of x = ————————.

10 Mode of values 5, 6, 7, 3, 4, 8, 2, 5,4, 5 is ————————.

(10 × 1 = 10 marks)

Part B

Answer any **eight** *from the following.*
Each question carries 2 *marks.*

11 Define Statistics.

12 What do you mean by Index Numbers ?

13 What is meant by Cartograms ?

14 If $U = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{4, 7, 10\}$ $B = \{2, 4, 5, 8\}$. Find $A' \cap B$.

15 Find the value of the determinant $\begin{vmatrix} 51 & 33 \\ 12 & 19 \end{vmatrix}$.

16 If $A = \begin{bmatrix} 44 & 22 \\ 33 & 55 \end{bmatrix}$. Find the minor of element 55.

17 Find the 7th term of series 1, 4, 7 ————————.

18 Solve $14x - 21 - 3x + 13 = 41 + 6x - 19$.

19 Find median for the following values :

87, 54, 63, 12, 78, 34, 36, 98

20 Find the Range and Co-efficient of Range for the following values :

65, 29, 98, 15,7, 8, 11,7, 43.

(8 × 2 = 16 marks)

Part C (Short Essay Questions)

Answer any **six** *questions from the following.*
Each question carries 4 *marks.*

21 What are the essentials of a good questionnaire ?

22 What are the important functions of statistics ?

23 In a factory there are 50 workers, 30 can speak in Hindi, 25 in Malayalam and 15 can speak in both the languages. Find the number of people who can speak at least one of the languages. How many cannot speak in any of these languages ?

24 Compute 3 yearly moving average from the following data :

Year	:	2015	2016	2017	2018	2019
Sales (in 000 units)	:	130	141	147	167	189

25 Calculate Arithmetic mean from the following data :

Values	:	5	15	25	35	45	55	65	75
Frequency	:	10	14	44	35	23	16	12	5

26 Compute Mean Deviation about mean from the following data :

Values	:	1	3	5	7	9	11
Frequency	:	2	4	6	4	2	6

27 Solve the equation $x^2 + x - 12 = 0$.

28 Calculate simple index number from the following data :

Commodities	Price in 2015	Price in 2020
A	5	8
B	4	6
C	2	4
D	8	11
E	13	17

(6 × 4 = 24 marks)

Turn over

Part D (Essay Questions)

Answer any **two** *questions from the following.*
Each question carries 15 *marks.*

29 What is statistical enquiry ? What are the different stages of statistical enquiry ?

30 Solve the following equations by using matrix :

$2x - 3y = 14$
$3x + 4y = 4.$

31 Calculate Standard Deviation and coefficient of variation from the following values :

Size	:	1	3	5	7	9	11	13	15
Frequency	:	2	4	5	7	9	8	4	2

(2 × 15 = 30 marks)

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B.Com./B.B.A.

BCM 3A 11/BBA 3A 11—BASIC NUMERICAL SKILLS

(2017–2018 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes	**Total No. of Questions : 20**	**Maximum : 20 Marks**

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

BCM 3A 11/BBA 3A 11—BASIC NUMERICAL SKILLS

(Multiple Choice Questions for SDE Candidates)

1. The Value of the determinant $\begin{vmatrix} 5 & 6 \\ 3 & 4 \end{vmatrix}$ is ————————.

   (A) 2. (B) – 2.

   (C) 38. (D) – 38.

2. Quadratic eqn $ax^2 + bx + c = 0$ has equal roots if :

   (A) $b - 4ac < 0$. (B) $b - 4ac > 0$.

   (C) $b - 4ac = 0$. (D) $b - 4ac = 1$.

3. ———————— are mathematical statements that contains one or more derivatives.

   (A) Sets. (B) Linear eqn.

   (C) Equation. (D) None.

4. The formula used for finding the roots of a quadratics eqn is known as :

   (A) Completing the square. (B) Factoring.

   (C) Quadratic formula. (D) None.

5. R (x) = = C (x) denotes :

   (A) Profit function. (B) Market equilibrium.

   (C) BEP. (D) None.

6. Find the 7th term of series 1,4,7 ————————.

   (A) 22. (B) 19.

   (C) 16. (D) 25.

7. If 2, 5, 8 ———————— is A.P the t 20 is :

   (A) 60. (B) 59.

   (C) 58. (D) None.

8. Given the term in the sequence 1, 3, 7, 15, 31 next is _____.

(A) 62. (B) 63.

(C) 46. (D) 66.

9. If $2, x, 8$ are the successive terms of a GP.the value of x is :

(A) 5. (B) 4.

(C) -4. (D) ± 4.

10. The sum at the end of 2 years for 1000 at 10% p.a. compounded yearly :

(A) 100. (B) 210.

(C) 1100. (D) 1210.

11. Statistics is applied in :

(A) Economics. (B) Business management.

(C) Commerce and Industry. (D) All these.

12. A frequency distribution can be :

(A) Dicrete. (B) Continuous.

(C) Neither. (D) Either.

13. The process of arranging data in groups according to similarities in character is called :

(A) Tabulation. (B) Classification.

(C) Tabulation or classifaction. (D) None.

14. The graphical representation of a cumulative frequency distribution is called :

(A) Histogram. (B) Ogive.

(C) Both. (D) None.

15. The point of intersection of the less than and the greater than ogives corresponds to :

(A) Mean. (B) Mode.

(C) Median. (D) Geometric mean.

16. In chronological classification data are classified on the basis of :

(A) Attributes. (B) Class intervals.

(C) Time. (D) Area.

17. ———— is the sum of the values divided by the total number of items in the set.

(A) Mode. (B) Median.

(C) Mean. (D) Skewness.

18. A time series is a set of data recorded :

(A) Periodically. (B) At time intervals.

(C) At successive points of time. (D) All the above.

19. ———— Index numbere are

(A) Special type of averages. (B) Measure of the economic barometers.

(C) Measure of the relative changes. (D) All of these.

20. ———— variations are periodic movements.

(A) Seasonal. (B) Secular trend.

(C) Cyclic. (D) Irregular.